

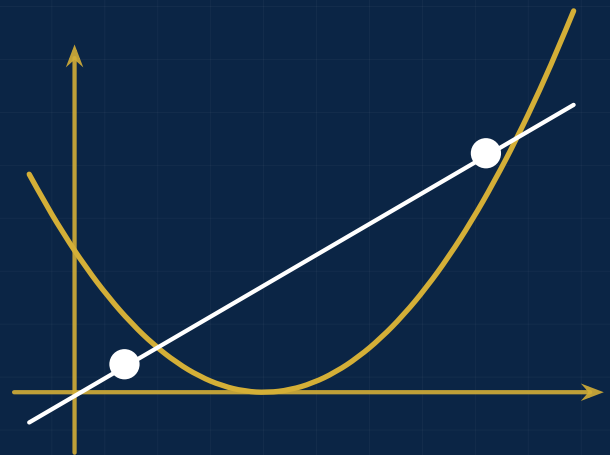


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 5

# Graphs of Functions

Table, plot, smooth curve. Then read the curve: solve equations with a line, name a shape from its equation, sketch a quadratic, decode a trig graph.



5

VARIANTS

40

PAST-PAPER PARTS

36

PRACTICE QUESTIONS

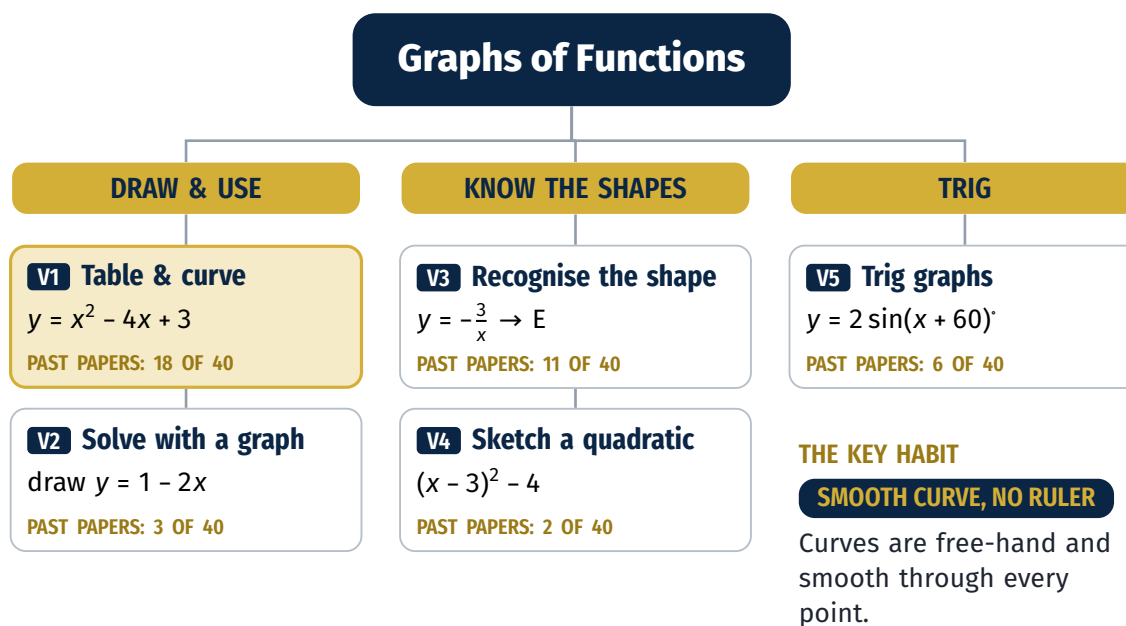
## YOU WILL MASTER

- V1 Table & curve
- V2 Solve with a graph
- V3 Recognise the shape
- V4 Sketch a quadratic
- V5 Trig graphs

## TOPIC MAP • EVERY VARIANT ON ONE PAGE

# Graphs of functions at a glance

 Your notes



## ★ How often does each variant come up?

22 past-paper questions (Jan 2020 → Nov 2025) hold 40 parts; 36 practice parts — grouped by type.

|                               | PAST PAPERS (OUT OF 40) | PRACTICE PARTS |
|-------------------------------|-------------------------|----------------|
| <b>V1</b> Table, curve        | 18                      | 13             |
| <b>V2</b> Solve with a graph  | 3                       | 7              |
| <b>V3</b> Recognise the shape | 11                      | 3              |
| <b>V4</b> Sketch a quadratic  | 2                       | 9              |
| <b>V5</b> Trig graphs         | 6                       | 4              |

**WHAT THIS TELLS YOU** A “complete the table, draw the graph” question is on almost every paper (4 marks, early). A “which graph is which” question (3 marks) is the next most common. Both are pure technique.

## BEFORE YOU START

## The big idea

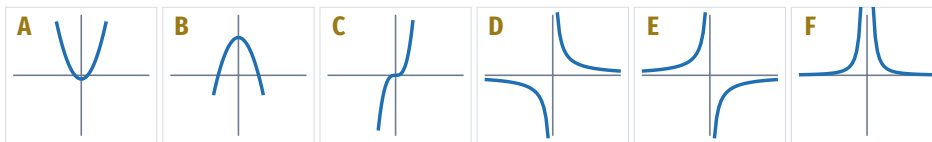
**Every point on a graph solves its equation.**

$(2, -1)$  is on  $y = x^2 - 4x + 3$  because  $4 - 8 + 3 = -1$ .

So: a **table** gives points to draw; the graph gives **solutions** back; and each equation type has its own **shape**.



## Know the shapes



A  $y = x^2$  (U)   B  $y = -x^2$  (∩)   C  $y = x^3$    D  $y = \frac{1}{x}$    E  $y = -\frac{1}{x}$    F  $y = \frac{1}{x^2}$



## CALCULATOR FIRST, THEN CHECK

Table mode:  $f(x) = x^2 - 4x + 3$ , start -2, end 4, step 1 — the whole table at once. Check one value by hand.

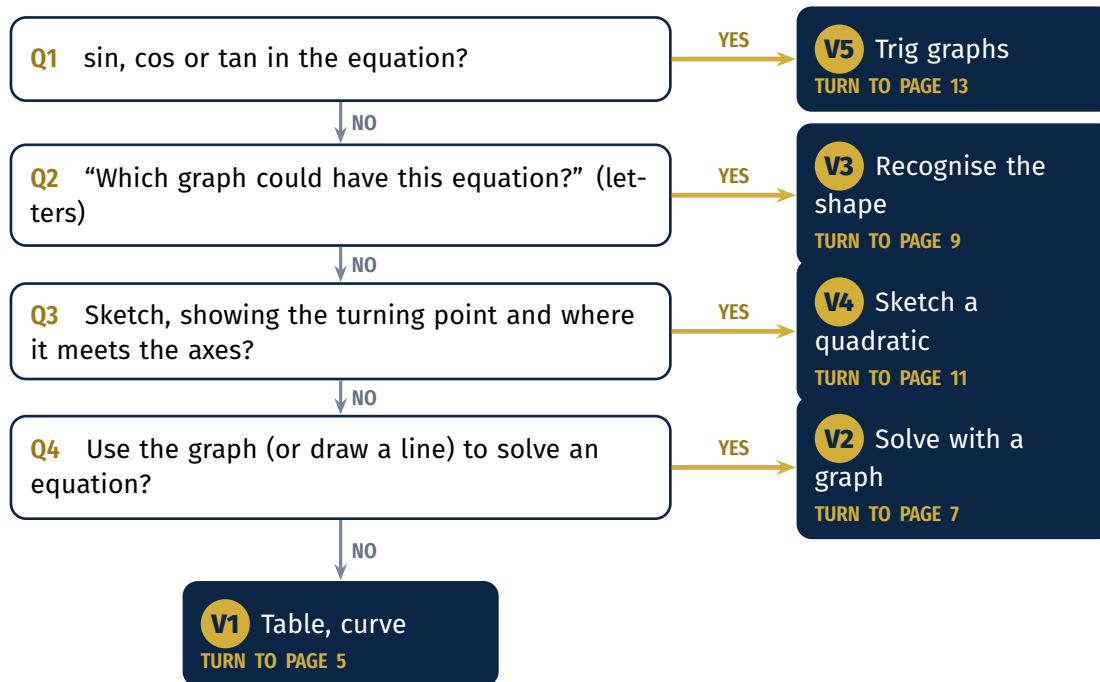
 Your notes

## COUNT / RECOGNISE, THEN CHOOSE

# Which variant am I?



Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

**Blue** = letter terms  $4x$ ,  $-3y$

**Orange** = plain numbers  $17$ ,  $-13$

**STEP 1** **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

**ON EVERY VARIANT SPREAD**

What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1 MOST COMMON IN THE EXAM

## Table &amp; Curve

Work out each  $y$  carefully, plot, and join with one smooth curve.

SEEN IN PAST PAPERS

18 / 40

PRACTICE QUESTIONS: 13



Your notes

**Came up in:** Jan 2020 P2HR Q15 • May 2021 P2H Q15 • Jan 2022 P1HR Q4 • Jan 2022 P2HR Q11 • Jun 2023 P2H Q4 • Nov 2023 P1H Q12 • Nov 2024 P1H Q11 • Jun 2025 P1H Q10 • Nov 2025 P2H Q12

## ? What is it?

### In plain words

Fill the table: substitute each  $x$  —

**bracket negatives:**  $(-2)^2 = 4$ .

Plot each point with a small cross and join with **one smooth curve** (no ruler, no straight bits). A U-curve has a **rounded bottom**, not a point.

### IT LOOKS LIKE

$$y = x^2 - 4x + 3$$

$$y = \frac{6}{x}$$

$$y = x^3 - 3x + 1$$

$$x = -2: (-2)^2 - 4(-2) + 3$$

$$= 4 + 8 + 3 = 15$$

$$\text{not } -4 + 8 + 3 \times$$

BRACKET THE NEGATIVE



## How do I do it?

### 1 Table

Substitute each  $x$ ; bracket negatives (or use table mode).

### 2 Pattern

Quadratic tables are symmetric — use it to spot slips.

### 3 Plot

Small crosses, exactly.

### 4 Curve

One smooth curve through all points, over the full range.

### SMOOTH CURVE, NO RULER

A point off the curve: recheck the table.

**V1 Table & Curve**

VARIANT 1 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

“Complete the table of values for  $y = x^2 - 4x + 3$ ”

→ **Substitute; bracket negatives**

“On the grid, draw the graph ... for values of  $x$  from  $-2$  to  $4$ ”

→ **Plot; smooth curve, whole range**

$y = \frac{6}{x}$ ,  $y = \frac{1}{x}(x^2 + 4)$

→ **Reciprocal: never touches the axes****WORKED EXAMPLE****PAST PAPER**

January 2022 · Paper 1HR · Q4 · 4 marks

(a) Complete the table of values for  $y = x^2 - 4x + 3$  for  $x = -2, -1, 0, 1, 2, 3, 4$ . (b) Draw the graph for  $x$  from  $-2$  to  $4$ .

**STEP 1**  $x = -2 : 4+8+3 = 15$ ;  $x = 1 : 1-4+3 = 0$  ◀ substitute

**STEP 2**  $y : 15, 8, 3, 0, -1, 0, 3$  ◀ (a) the table

**STEP 3** symmetric about  $x = 2$  ◀ check the pattern

**STEP 4** Plot the 7 points; one smooth U-curve, lowest at  $(2, -1)$  ◀ (b)

**ANSWER** 15, 8, 3, 0, -1, 0, 3; U-curve, minimum  $(2, -1)$  ✓  
symmetric ✓

**WATCH OUT**

✗ a pointed V at the bottom

✓ a rounded U

The curve turns smoothly between  $x = 1$  and  $x = 3$ .

**EXAM TIP**

Turn the paper and draw from the inside of your wrist — smoother curves.

**Check yourself**

calculator first · answers on page 15

**1.1**

Jun 2023 P2H

$y = x^2 - x - 4$ . Find  $y$  for  $x = -3, -1, 0, 2, 3$ .

Answer: .....

**1.2**

Jan 2022 P2HR

$y = \frac{6}{x}$ . Find  $y$  for  $x = 0.5, 2, 4, 5$ .

Answer: .....

**1.3**

Nov 2023 P1H

$y = x^3 - 3x + 1$ . Find  $y$  for  $x = -2, -1, 0, 2$ .

Answer: .....

**V2**

VARIANT 2

# Solve Equations with a Graph

Rearrange the new equation into “curve = line”; draw the line; read the x values where they cross.

SEEN IN PAST PAPERS

3 / 40

PRACTICE QUESTIONS: 7


  
Your notes

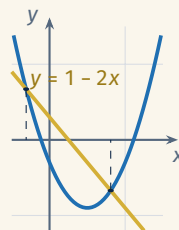
Came up in: Jan 2022 P2H Q21 • Nov 2025 P2H Q12(c)

## ? What is it?

### In plain words

Solutions of “curve = 0” are where the curve crosses the **x-axis**.

For a different equation, make the curve’s expression appear on one side; the other side is a **straight line** to draw. The answers are the **x-values** of the crossings.



### IT LOOKS LIKE

$$2x^2 - 4x - 1 = 0$$

$$x + \frac{4}{x} = 6$$

## How do I do it?

**1 Match**

Rearrange so one side is the curve you have.

**2 Line**

The other side is the line to draw.

**3 Draw**

Rule the line on the grid.

**4 Read**

x-values of the crossings, to 1 d.p.

**CURVE = LINE**

Answers are x-values only.

**V2 Solve Equations with a Graph**

VARIANT 2 • SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

“Use the graph to find estimates for the solutions of  $2x^2 - 4x - 1 = 0$ ”

→ **Where the curve crosses  $y = 0$** 

“By drawing a suitable straight line ... solve  $x^2 - x - 1 = 0$ ”

→ **Make  $2x^2 - 4x - 1$  appear**

“correct to one decimal place”

→ **Read carefully on the small squares****WORKED EXAMPLE** **PAST PAPER**

January 2022 • Paper 2H • Q21(b) • 3 marks

Part of the graph of  $y = 2x^2 - 4x - 1$  is shown. By drawing a suitable straight line, find estimates for the solutions of  $x^2 - x - 1 = 0$ .

**STEP 1**  $2x^2 - 2x - 2 = 0$ ◀  $\times 2$  to get  $2x^2$ **STEP 2**  $2x^2 - 4x - 1 = -2x + 1$ 

◀ curve on the left

**STEP 3** draw  $y = 1 - 2x$ 

◀ the line

**STEP 4**  $x \approx -0.6, \quad x \approx 1.6$ 

◀ read the crossings

**ANSWER**  $x \approx -0.6$  and  $x \approx 1.6$ ✓  $1.6^2 - 1.6 - 1 = -0.04 \approx 0$  ✓**WATCH OUT**✗ giving the  $y$ -values✓ give the  $x$ -values

The solutions are the  $x$ -coordinates of the crossings.

**EXAM TIP**

Write the rearrangement line — it is the method mark, even if your reading is a little off.

**Check yourself**

calculator first • answers on page 15

**2.1**

Jan 2022 P2H

Use the graph of  $y = 2x^2 - 4x - 1$  to solve  $2x^2 - 4x - 1 = 0$  (1 d.p.).

Answer: .....

**2.2**

Nov 2025 P2H

The graph of  $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$  is drawn. Which line solves  $x + \frac{4}{x} = 6$ ? Solutions?

Answer: .....

**2.3**

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The graph of  $y = x^2 - 3x$  is drawn. Which line would you draw to solve  $x^2 - 4x + 1 = 0$ ?

Answer: .....

**V3**

VARIANT 3

**Recognise the Shape**

Read the highest power of  $x$  and the sign — then match the shape.

SEEN IN PAST PAPERS

11 / 40

PRACTICE QUESTIONS: 3



Your notes

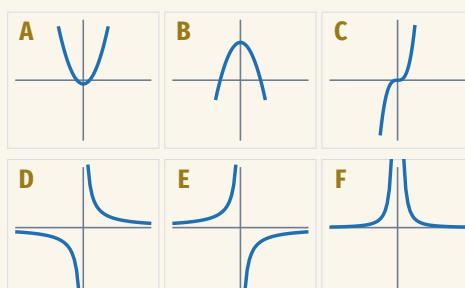
**Came up in:** Jan 2020 P2H Q15 • Jun 2022 P2HR Q16 • Jun 2023 P2HR Q12 • Jun 2023 P2H Q18 • Nov 2023 P1H Q12(c) • Jun 2024 P2HR Q16 • Jun 2025 P2H Q14

**? What is it?****In plain words**

$x^2$ : **U** (or **n** if negative).

$x^3$ : **S-shape**.  $\frac{1}{x}$ : two pieces in opposite corners.  $\frac{1}{x^2}$ : two pieces, **both above**.

A minus sign **flips** the shape.

**How do I do it?****1 Power**

Highest power: 1 line, 2 parabola, 3 cubic.

**2 Bottom?**

$x$  on the bottom: reciprocal (never touches the axes).

**3 Sign**

Negative in front: flip upside down.

**4 Check**

Put  $x = 0$  (or  $x = 1$ ) in: does the graph agree?

**POWER, BOTTOM, SIGN**

Three questions name any shape.

**V3** Recognise the Shape

VARIANT 3 • SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

“Write down the letter of the graph that could have the equation”

→ Power, bottom, sign

$$y = -\frac{3}{x}$$

→ Reciprocal, flipped: top-left and bottom-right

$$y = \tan x^\circ, y = \sin x^\circ$$

→ Trig shapes (V5)

**WORKED EXAMPLE****PAST PAPER**

June 2023 • Paper 2HR • Q12 (shapes as in the gallery) • 3

marks

Match each equation to a shape in the gallery: (a)  $y = \frac{10}{x^2}$  (b)  $y = -\frac{3}{x}$   
(c)  $y = 5 - x^2$ .

**STEP 1**  $\frac{10}{x^2} > 0$  always

◀ (a) two pieces, both above: F

**STEP 2**  $-\frac{3}{x} : x > 0 \Rightarrow y < 0$

◀ (b) reciprocal flipped: E

**STEP 3**  $-x^2 + 5$

◀ (c) negative  $x^2$ : n, B**ANSWER** (a) F (b) E (c) B✓  $x = 1$ : 10, -3, 4 ✓**WATCH OUT**

✗  $y = -\frac{3}{x}$  is D

✓ E

The minus flips it into the other two corners.

**EXAM TIP**

Test one value:  $x = 1$  in  $y = -\frac{3}{x}$  gives -3 — the graph must be below the axis there.

**Check yourself**

calculator first • answers on page 15

**3.1**

Jun 2025 P2H

Which gallery shape could be  $y = 2x^3$ ?

Answer: .....

**3.2**

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Which gallery shape could be  $y = \frac{4}{x}$ ?

Answer: .....

**3.3**

Dr Eslam

Which gallery shape could be  $y = x^2 - 4$ ?

Answer: .....

**V4**

VARIANT 4

**Sketch a Quadratic**

Complete the square for the turning point;  
 $x = 0$  for the  $y$ -intercept;  $y = 0$  for the roots.

SEEN IN PAST PAPERS

2 / 40

PRACTICE QUESTIONS: 9


  
Your notes

Came up in: Jan 2020 P1H Q20 • Nov 2024 P1H Q23

**? What is it?****In plain words**

A **sketch** is not a plot: show the shape and the **key points** only.

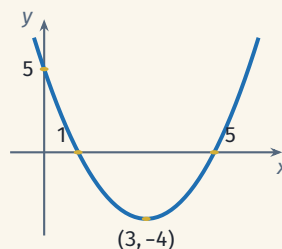
$y = (x - 3)^2 - 4$ : turning point  $(3, -4)$ .

$x = 0$ : **y-intercept**.  $y = 0$ : **roots**.

**IT LOOKS LIKE**

$$y = x^2 - 6x + 5$$

$$y = 4(x - 1)^2 - a$$

**How do I do it?****1 Turning point**

Complete the square:  $(x - p)^2 + q \rightarrow (p, q)$ .

**2 y-axis**

Put  $x = 0$ .

**3 x-axis**

Put  $y = 0$ ; factorise or solve.

**4 Sketch**

U (or n) through the points; label them.

**TP • x = 0 • y = 0** Label every point you found.

**V4 Sketch a Quadratic**

VARIANT 4 · SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"Sketch the curve, showing the turning point" → **Complete the square**

"points of intersection with the coordinate axes" →  **$x = 0$  and  $y = 0$**

"The line  $y = k$  meets the curve at  $(p, k)$ ,  $(q, k)$ " →  **$p, q$  are roots of curve =  $k$**

**WORKED EXAMPLE DR ESAM**

Sketch · 4 marks

Sketch  $y = x^2 - 6x + 5$ , showing the turning point and where it meets the axes.

**STEP 1**  $x^2 - 6x + 5 = (x - 3)^2 - 4$  ◀ complete the square

**STEP 2** TP (3, -4) ◀ minimum (positive  $x^2$ )

**STEP 3**  $x = 0 : (0, 5)$  ◀ y-intercept

**STEP 4**  $(x - 1)(x - 5) = 0 : (1, 0), (5, 0)$  ◀ roots

**ANSWER** U-curve: TP (3, -4), (0, 5), (1, 0), (5, 0) ✓ TP halfway between the roots ✓

**WATCH OUT**

× TP (-3, -4)

✓ (3, -4)

$(x - 3)^2$  is smallest at  $x = +3$ .

**EXAM TIP**

The turning point sits exactly halfway between the roots — a free check.

**Check yourself**

calculator first · answers on page 15

**4.1**

Jan 2020 P1H

$y = 4(x - 1)^2 - a$ ,  $a > 4$ .  
Give the turning point  
and the y-intercept.

Answer: .....

**4.2**

Nov 2024 P1H

$y = x^2 - 8x - 9$  meets  
 $y = k$  at  $(p, k)$  and  $(q, k)$ ,  
 $p - q = 14$ . Find  $k$ .

Answer: .....

**4.3**

Dr Eslam

Find the turning point  
of  $y = x^2 - 4x + 1$ .

Answer: .....

**V5**

VARIANT 5

**Trig Graphs**

Know sin, cos, tan by heart; then read the stretch ( $a$ ), the shift ( $b$ ) and the lift ( $c$ ).

SEEN IN PAST PAPERS

6 / 40

PRACTICE QUESTIONS: 4



Your notes

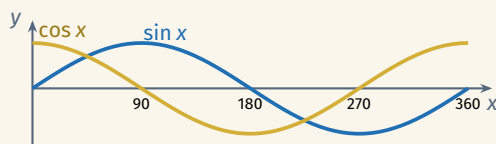
Came up in: Jun 2022 P2H Q23 • Jun 2023 P2HR Q26 • Jun 2024 P2H Q25 • Jun 2025 P2HR Q25

## ? What is it?

### In plain words

$\sin x$ : starts at 0, top at  $90^\circ$ .  $\cos x$ : starts at 1, bottom at  $180^\circ$ . Both repeat every  $360^\circ$ .

$y = a \sin(x + b) + c$ :  $a$  = height above the middle;  $c$  = middle line;  $b$  slides it left.  $\cos bx$  repeats every  $\frac{360}{b}$ .



### IT LOOKS LIKE

$$2 \sin(x + 60^\circ)$$

$$\cos\left(\frac{x}{2}\right)$$

$$a \cos bx + c$$



## How do I do it?

### 1 Middle $c$

$$\frac{\max + \min}{2}$$

### 2 Height $a$

$$\frac{\max - \min}{2}$$

### 3 Period / shift

Period  $\frac{360}{b}$ ; or where the peak moved.

### 4 Points

Peak where the bracket = 90 (sin) or 0 (cos).

### MIDDLE, HEIGHT, SHIFT

Max and min give  $a$  and  $c$ .

**V5 Trig Graphs**

VARIANT 5 • SEE IT DONE, THEN TRY


  
Your notes
**WHEN YOU SEE → YOU DO**

"Find the coordinates of A" (a peak) → **Bracket = 90° for sin**

"Find a, b and c" → **c middle, a half the height, b from the period or shift**

$\cos\left(\frac{x}{2}\right)$  → **Stretched: repeats every 720°**

**WORKED EXAMPLE** **PAST PAPER**

June 2024 • Paper 2H • Q25 • 2 marks

The graph of  $y = 2 \sin(x + 60)^\circ$  is sketched. A is the first maximum point; B is where the curve next crosses the x-axis going down. Find A and B.

**STEP 1**  $x + 60 = 90 \Rightarrow x = 30$  ◀ sin peaks at 90°

**STEP 2** A(30, 2) ◀ height a = 2

**STEP 3**  $x + 60 = 180 \Rightarrow x = 120$  ◀ sin is 0 again at 180°

**STEP 4** B(120, 0) ◀ on the axis

**ANSWER** A(30, 2), B(120, 0)✓  $2 \sin 180^\circ = 0$  ✓**WATCH OUT**✗  $x + 60 = 90$  so  $x = 150$ ✓  $x = 30$ 

Subtract the 60, don't add it: the graph moved left.

**EXAM TIP**

Write the bracket = 90 or = 180 first — the coordinate follows.

**Check yourself**

calculator first • answers on page 15

**5.1** Jun 2022 P2H

$y = \cos\left(\frac{x}{2}\right)$ : A first crossing of the x-axis, B the minimum. Find A and B.

Answer: .....

**5.2** Jun 2023 P2HR

$y = a \cos bx^\circ + c$ : max 3, min -1, peaks every 90°. Find a, b, c.

Answer: .....

**5.3** Jun 2025 P2HR

$y = a \sin(x + b)^\circ + c$ : max 4 at  $x = 45$ , min -2. Find a, b, c.

Answer: .....

## CHECK YOURSELF

# Answers



Every answer below was checked with sympy (check\_answers.py).

**1.1** 8, -2, -4, -2, 2

**1.2** 12, 3, 1.5, 1.2

**1.3** -1, 3, 1, 3

**2.1**  $x \approx -0.2, 2.2$

**2.2**  $y = 3$ ;  $x \approx 0.8, 5.2$

**2.3**  $y = x - 1$

**3.1** C

**3.2** D

**3.3** A

**4.1** TP  $(1, -a)$ ;  $(0, 4 - a)$

**4.2**  $k = 24$

**4.3**  $(2, -3)$

**5.1**  $A(180, 0)$ ,  $B(360, -1)$

**5.2**  $a = 2$ ,  $b = 4$ ,  $c = 1$

**5.3**  $a = 3$ ,  $b = 45$ ,  $c = 1$

## THE WHOLE TOPIC ON ONE PAGE

# Graphs cheat sheet


  
Your notes
**V1 Table, curve 18/40**

$$(-2)^2 = 4$$

Bracket negatives; one smooth curve.

**V2 Solve 3/40**

$$2x^2 - 4x - 1 = 1 - 2x$$

Curve = line; read x.

**V3 Shapes 11/40**

$$-\frac{3}{x} \rightarrow E$$

Power, bottom, sign.

**V4 Sketch 2/40**

$$(x - 3)^2 - 4$$

TP,  $x = 0$ ,  $y = 0$ .**V5 Trig 6/40**

$$a \sin(x + b) + c$$

Middle, height, shift.

**TOP 3 LOST MARKS**

1.  $-2^2 = -4$  in a table ×
2. ruled or pointed curves ×
3. giving y for the solutions ×

**BOTTOM LINE** Bracket negatives • smooth curve • curve = line • power, bottom, sign.